LabTech Testing Limited

STANDARD TEST PROCEDURE

HKCI-TM3
Determination of Geometry of
Vertical Shaft Excavation by Ultrasonic Echo Sounding

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Amendment Ledger

Issue No.	Page (s)	Revision Outline
2	1	C.1.2 add the full name of TM3
	3	C.3.2 specify water as the propagation medium
	4	C.4.3 change bentonite slurry solutions to water
	4	C.4.4 specify the temperature for normal operation of UES equipment
	5	C.5.2.2 change ” to °
	5	C.5.3.1 change 100 kHz to 110 kHz
	6-7	Section 6 revise completely in order to match the testing procedure on site
	7	C.7.2 amend the equation of calculation of verticality
	8	Section 8 change from Report to Record
	8	Section 9 add as Report
	10	C.4 change frequency of calibration to once per year
	10	C.5 specify the calibration and check requirement of caliper
3	7	C.7.1 change Scaling factor to Scaling factor from valid calibration certificate
4	6	C.5.5 add ANNEX II for handling of equipment
	6	C.6.6 specify two measurements of internal diameter of casting
	7	C. 6.21 to 6.23 specify conditions to be communicated with customer
	8	C.8.3 add the form daily record
	10	ANNEX I C.3 align the criteria of UES check
	11	ANNEX II new requirements of handling and maintenance of equipment

REVISION LEDGER

Rev. No.	Section No.	Page(s)	Revision Outline
5	All	All	Change Issue No. to Revision No.
6	All	All	Change excavation holes/shaft to vertical shaft excavation
	1	4	Revise C1.1 and C1.2
	2	4	Revise C2.2 and C2.3
	2	4	C.2.3 & C.2.4 are deleted
	3	4	C3.3 change 'boundaries' to 'interfaces'
	4	5	Revise C4.2 and C4.3
	5	5	C5.1.1 change 'diameter' to 'space'
	5	5	C5.2.1 add 'at least ± 0.5% accuracy'
	5	6	Revise C5.2.2
	5	6	Add C5.2.3 to C5.2.5
	5	6	C5.3.1 add 'The ultrasonic sensor head consists of a transmitter(s) and a receiver(s)'.
	5	7	C5.4.2 add 'The equipment has adjustable gain to optimize detection of the transmitted pulse by the ultrasonic sensor head under test'
	5	7	Add C5.4.4
	6	7	C6.5 add 'measure the temperature of the water/bentonite slurry sample'
	7	9-10	Revise C7.1 and C7.2
	9	10-11	C9.1 add 'time' and 'type and temperature of transmission medium'
	9	11	C9.2 change 'diameter' to 'size'
	9	11	C9.3 add 'without any cuttings of the profiles'
	Annex I	12	Add 6. Verification of echo sounder to known profiles and fluid properties
7	Annex I	12	5 Change frequency of caliper to once per year

1 Introduction

1.1 This test method covers calibration of measuring apparatus and determination of field procedures for measuring the profile of a vertical shaft excavation, such as dimension of a bell-out and verticality of an excavation using Ultrasonic Echo Sounding (UES) technique.

1.2 The test method is prepared based on HKCI : TM3 Issue 2 (September 2024) 'Determination of Geometry of Vertical Shaft Excavation by Ultrasonic Echo Sounding'.

2 Definitions

2.1 UES: Ultrasonic Echo Sounding.

2.2 UES Profile: A graphical record of measured distances against depth.

2.3 Scaling factor: A factor used to convert the values measured on an UES profile into horizontal dimensions of the tested shaft excavation. A scaling factor shall be determined with the use of two first- arrival-time signals in each test.

2.4 Ultrasonic sensor head: a combination of transmitters and receivers of an UES equipment.

3 Principle

3.1 UES measurements involve measuring the propagation time of an ultrasonic pulse for a reflection off the sidewalls (such as steel casing, bedrock, soil layer, concrete guide wall, etc.) of a vertical shaft excavation.

3.2 The excavation should be filled with a propagation medium (such as water, bentonite slurry, etc.) so as to obtain a good coupling effect for ultrasonic wave propagation.

3.3 The transmitted ultrasonic pulses are reflected as a result of changes of the acoustic impedance at different material interfaces.

3.4 Each interface of two different materials (i.e. difference in acoustic impedance) produces partial reflection (i.e. receiving signals) and transmission to occur at normal incidence, and even makes the receiving signals vary.

4 Limitations

4.1 Due to the design of the instrumentation of the UES equipment, only two directions (i.e. X-X' direction and Y-Y' direction) can be measured at any one time. The testing area is limited to two directions and the conditions of drilled shafts in other elevations are not covered.

4.2 In performing an UES test, the sensor head of the UES equipment is required to be placed in the centre of an excavation to produce reliable UES profiles at an optimal position. However, for inclined shafts, a number of measurements should be carried out to obtain an actual picture of these inclined shafts.

4.3 The transmission medium in diaphragm walls is sometimes filled with bentonite slurry solutions. However, the UES equipment may not be able to measure detect the sidewalls in muddy water with sufficient clarity. This is due to the high attenuation (including scattering, absorption and reflection) of ultrasonic waves in muddy water. As a result, clear reflecting signals cannot be obtained on UES profiles even though the UES equipment is set in a high signal-to-noise ratio setting.

4.4 The temperature for normal operation of UES equipment should be ranged from -10 to 50°C.

5 Equipment

5.1 Equipment for allowing excavation holes/shafts inspection

5.1.1 Excavation holes/shafts: to provide access for an ultrasonic sensor head of the UES equipment, steel casings, if any, are installed during the drilling of a vertical excavation. The internal space of the excavation holes/shafts should be sufficient to allow the easy passage of the ultrasonic sensor head over the entire excavation depth. A vertical shaft excavation typically has an internal diameter from 1 to 8 m, and should be filled with water/bentonite slurry to provide a propagation medium for ultrasonic pulse propagation.

5.2 Equipment for determining physical test parameters

5.2.1 Weighted measuring tape: a plumb bob connected to a measuring tape with at least ± 0.5% accuracy should be used as a dummy ultrasonic sensor head to check free

passage through and determine the unobstructed length of a vertical shaft excavation.

5.2.2 Compass or gyroscope – used for determining direction relative to the Earth's magnetic poles at the test excavation.

5.2.3 Measuring tape – with at least ± 0.5% accuracy, used for on-site measuring horizontal dimensions of a vertical shaft excavation.

5.2.4 Temperature measuring device – used for measuring the temperature of the water/bentonite slurry solution in the excavation to ensure that the temperature of the propagation medium does not exceed the allowable temperature value as recommended by the manufacturer of the UES equipment.

5.2.5 Spirit leveller – used to ensure an UES equipment in a horizontal position prior to testing.

5.3 Equipment for obtaining measurements

5.3.1 Ultrasonic sensor head: The ultrasonic sensor head consists of a transmitter(s) and a receiver(s). The transmitters can generate an ultrasonic pulse with a frequency between 80 to 110 kHz.

5.3.2 Signal transmission cable: the signal cable used to deploy an ultrasonic sensor head and transmit data from the sensor head is sufficient robust to support the sensor head's weight. The cable is abrasion resistant to allow repeated field use and maintain flexibility in the range of anticipated temperatures. All cable connector or splices, if any, are watertight. Where the signal transmission cables exit the sensor head, suitable cable guides, pulleys or cushioning material are fitted with the UES equipment to minimize abrasion and generally assist with smooth deployment of the ultrasonic sensor head. The total length of cable should cover the measured depth, and typically at least 100m.

5.4 Equipment for recording and displaying data

5.4.1 General: the signals from the transmitters and the receivers of the ultrasonic sensor head should be transmitted to a field rugged equipment for recording and displaying the data in the form of an ultrasonic profile.

5.4.2 Recording equipment: each transmitted ultrasonic pulse should immediately start the acquisition for the corresponding receiver. Analog signals of an ultrasonic pulse measured by the receivers are digitized by an analog to digital converter with a minimum sampling frequency of 4 Hz. The equipment has adjustable gain to optimize detection of the transmitted pulse by the ultrasonic sensor head under test.

5.4.3 Equipment for display of measured data: the equipment is capable of displaying the ranges of the sidewalls of a vertical shaft excavation in two orthogonal directions at any one time, and also record test information such as date and time of the measurement carried out, and depth of a vertical shaft excavation.

5.4.4 Apparatus for measuring value(s) on an UES profile – A calibrated caliper with at least ± 0.5% accuracy is used for measuring value(s) on an UES profile for determination of horizontal and vertical dimensions of a vertical shaft excavation.

5.5 Calibration and verification check requirements; and handling of equipment used are given in ANNEX I and ANNEX II respectively.

6 Testing Procedures

6.1 Check that the equipment is functioning correctly prior to stabilizing to site.

6.2 Check the ultrasonic sensor head to ensure it is clean and free from contaminants.

6.3 Confirm with the customer on-site the drawing and related information. Record the test locations and details of the pile onto worksheet.

6.4 Record the weather and regional temperature of the site by accessing HKO website.

6.5 Ensure that the tested shaft excavation(s) are filled with water/bentonite slurry at the date of measurement. Check the quality of water/bentonite slurry sample visually, measure the temperature of the water/bentonite slurry sample and check the water/bentonite slurry depth by dipmeter.

6.6 Measure the internal diameter of steel casing. Make two measurements at perpendicular directions (NS and WE).

6.7 Check the depth of the test vertical shaft excavation by a weighed measuring tape, record measurement directions by a magnetic compass, and any other specific

observation or given information (for example the presence of tremie pipe) relevant to each excavated hole/shaft tested that relate to the holes/shafts tested.

6.8 Set a crosshair by fishing line at the top of steel casing to locate the centre of the test excavated hole/shaft.

6.9 Remove any obstacle around the excavation to ensure that the test equipment can be securely operated above the excavated hole.

6.10 Set up the ultrasonic sensor head of the UES equipment nearly in the centre of the test excavated hole/shaft. Level the winch using a spirit level.

6.11 Re-check that the test equipment and the ultrasonic sensor head functioning correctly.

6.12 Check the paper making sure it is sufficient to complete the test.

6.13 Input the date, time, hole size, measurement range, measuring directions (X-X' & Y-Y'), up/down speed, paper feed rate, shift (%) into the controller.

6.14 Lower the ultrasonic sensor head in the water filled test excavated holes/shaft and verify that ultrasonic pulses are received in the recording equipment.

6.15 Perform a distance correction for adjusting reflected waves from the guide wall or the wall of casing to the position of the scale equal to the distance.

6.16 Select a proper range scale for the test excavated hole/shaft in the actual UES measurement.

6.17 Adjust the PAPER SPEED to 1:50 if the depth is less than 15m, 1:100 when the depth is between 15m to 40m and 1:200 when the depth is larger than 40m according to the user manual.

6.18 Start the actual UES measurement by carefully lower the ultrasonic sensor head down the test excavated hole/shaft at a steady rate ranged from 5m/min to 11m/min.

6.19 Set an appropriate signal gain during the actual UES measurement in order to increase signal-to-noise ratio to produce better results of receiving ultrasonic signals.

6.20 Check the quality of the UES profile obtained which shall be of good resolution and contrast. A typical UES profile is given in ANNEX III.

6.21 Before commencement of the measurement, inform the customer deviations such as

difference between the measured and nominal internal diameter of the steel casing. Termination or continuation of the test is up to the decision of the customer.

6.22 Confirm the test results and additional information to be reported (in case change of verticality is observed at toe) with customer before unplugging the UES equipment and demobilization.

6.23 Request the customer to confirm any decision or deviations in daily record.

7 Calculation

7.1 Test excavation holes/shafts dimension or "bell-out" dimension (L)

$$L = r * S_H$$

where

r is a horizontal value measured by a calibrated caliper on an UES profile at which a 'bell-out' is located (unit in mm); and

S_H is an average horizontal scaling factor (unit in m/mm) by taking at least three readings.

7.2 Verticality of the shaft excavation (V)

[The verticality is reported as a ratio (i.e. 1 : V)

For X-axis,

$$V_x = \frac{D}{|L_{x2} - L_{x1}|} = \frac{(D_2 - D_1)}{|r_{x2} - r_{x1}| * S_H}$$

For Y-axis,

$$V_Y = \frac{D}{|L_{Y2} - L_{Y1}|} = \frac{(D_2 - D_1)}{|r_{Y2} - r_{Y1}| * S_H}$$

For resultant verticality,

$$V = \frac{D}{\sqrt{|L_{x2} - L_{x1}|^2 + |L_{Y2} - L_{Y1}|^2}}$$

where

L_{x1} and L_{Y1} are the distances between the middle point (i.e. a half distance between two first-arrival-time on each side of the UES profile) and the reference line at the depth D_1 in X-axis and Y-axis respectively (unit in m);

L_{x2} and L_{Y2} are the distances between the middle point (i.e. a half distance between two first-arrival-time on each side of the UES profile) and the reference line at the depth D_2 in X-axis and Y-axis respectively (unit in m);

D_1 is a depth of a vertical shaft excavation at which $L_{x1}$and $L_{Y1}$are located (unit in m);

D_2 is a depth of a vertical shaft excavation at which $L_{x2}$and $L_{Y2}$are located (unit in m);

r_{x1} and r_{Y1} are horizontal values measured by a calibrated caliper on an UES profile at the depth D_1 in X-axis and Y-axis respectively
(unit in mm);

r_{Y2} and r_{Y2} are horizontal values measured by a calibrated caliper on an UES profile at the depth D_2 in X-axis and Y-axis respectively
(unit in mm); and

S_H is an average horizontal scaling factor (unit in m/mm) by taking at least three readings.

7.3 In addition to the overall verticality of a tested shaft excavation, successive evaluation of verticality at 10m (or 5m) interval is required.

8 Record

8.1 Worksheet: FST02-FDT-TM3

8.2 Report template: FDT-TM3-RT

8.3 Daily Record: FQS09

9 Report

9.1 The test report should contain as a minimum the following information:

- Name of customer, project title and contract no;
- Report no., date and time of test;
- Description of the testing UES equipment (4-channel in one measurement);

Identification of test staff and of person responsible for the validity of the test report; and

- Type and temperature of transmission medium.

9.2 Tested shaft excavation(s)

- Identification and location of test shaft excavation;
- As-built geometry of tested shaft excavation including nominal and/or actual size and length;
- Identification of directions of tested shaft excavation;
- Cut-off and ground elevation of tested shaft excavation, elevation of steel casing top, or length from casing top to the tested shaft excavation toe; and
- Any other specific observation or given information relevant to each tested shaft excavation (for example propagation media, tremie pipe inside the tested shaft excavation etc.) that relate to the shaft tested.

9.3 UES profiles: the UES profiles without any cuttings of the profiles should be presented logically and clearly for each shaft excavation tested and relevant information should accompany the profiles or be clearly associated with the profiles by cross-referencing, including the directions of the ultrasonic sensor head for the UES profile with time and depth axes and scales allowing clear data interpretation.

9.4 Determination of tested shaft excavation dimension and 'bell-out' dimension (if available): the tested shaft excavation and 'bell-out' dimensions are reported in unit of meter (m).

9.5 Calculation of verticality of a tested shaft excavation: the verticality is reported as a ratio (i.e. $1 : V$).

ANNEX I

Calibration and check requirements of equipment used for the test

1 The calibration requirements are based on HOKLAS Supplementary Criteria No. 2 and 16.

2 Calibration of designated frequency of UES equipment

- either by a 'competent calibration body' as defined in the Quality Manual or calibrate using reference oscilloscope, CRT, frequency analyzer or any reference measuring device; and
- frequency: once per 2 years.

3 Calibration of depth measuring device of UES equipment

- check against plumb-weighed measuring device before test on site. The Length of pile indicated on the UES profile shall not deviate more than 0.3m or 1% of the measured actual length of pile.

4 Calibration and check of steel rule and measuring tape

- calibrate against reference length standard of higher accuracy;
- a minimum of 5 points over the entire range;
- frequency: once per year; and
- check the general conditions every month.

5 Calibration and check of caliper

- by a 'competent calibration body' as defined in the Quality Manual;
- frequency: once per year; and
- check the general conditions every month.

6 Verification of echo sounder to known profiles and fluid properties

- by means of known profiles of different geometrical characteristics (e.g. distances, roughness, curvatures and inclinations, etc.) in water / various physical models and known fluid properties; and
- frequency: once per year.

ANNEX II

Handling and maintenance of equipment

1 Equipment taken to site measurement should be stored in designated container before carrying to site.

2 The UES

2.1 UES is stored in designated storage room in the laboratory with control access at the main gate.

2.2 The storage room is locked and the key of the lock is kept by laboratory staff.

2.3 As the UES can be operated within -10 to 50ºC, no specific requirements of environmental conditions of the storage warehouse are needed.

2.4 Within the store room, only tools and equipment are placed. No solvent or corrosive fluids would be stored as not to deteriorate the performance of the UES.

2.5 The UES should be checked for normal functions and then performance in accordance with clause 3 as stipulated in ANNEX I before test.

3 Plumb-weighed measuring tape, steel ruler and caliper

3.1 Keep the equipment in designated areas within the laboratory and keep away from solvent or corrosive fluids as not to deteriorate the equipment.

3.2 Check the conditions such as rust, clear marks, etc. before carrying to the site.

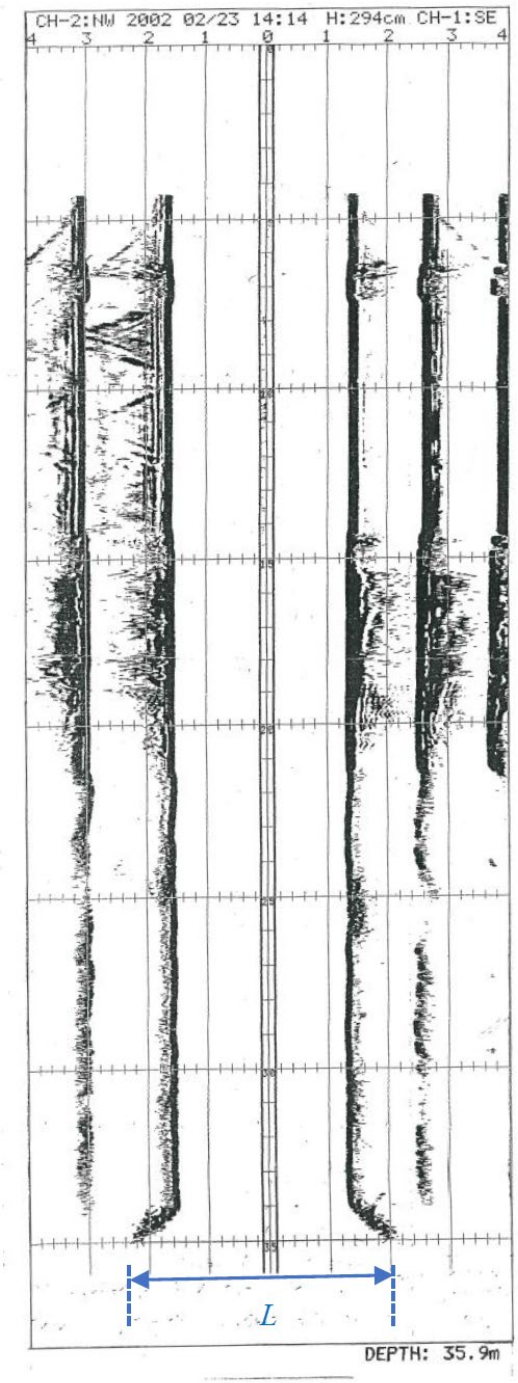
4 Magnetic compass

4.1 Keep the equipment in designated areas within the laboratory and keep away from solvent or corrosive fluids as not to deteriorate the equipment.

4.2 Check the general conditions before carrying to the site.

ANNEX III

Typical UES profile of a vertical shaft excavation



End of Procedure